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EEL4712 Digital Design

Lecture 35 – Metastability and Clock Domain Crossing

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Agenda

1. Motivation
2. Metastability & Clock Domain Crossing (CDC)
3. Solutions to Metastability
   - Dual flop synchronizer
   - Mux recirculation synchronizer
   - Handshake synchronizer (next lecture)
   - FIFO (next lecture)

Motivation

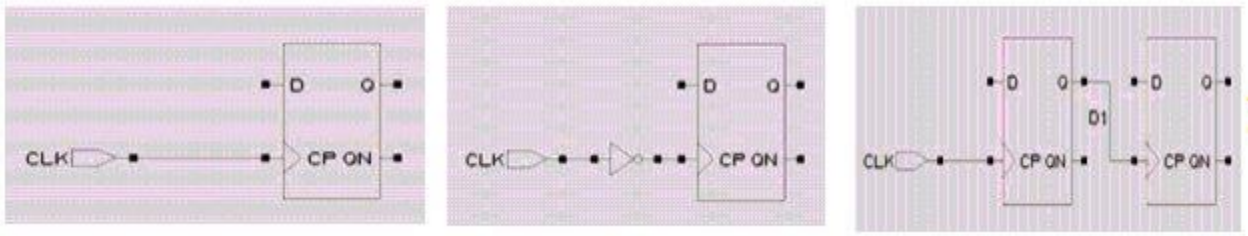
- What happens if:
  - A signal changes close at the clock edge?
- We assume:
  - Signals are 0 or 1
  - Flip-flops sample clean values
  - Timing constraints are met
- But what if they are not?
- A real system example
  - Button press-> FSM
  - FPGA – DRAM communication

Asynchronous Inputs Exist Everywhere

- Asynchronous Inputs
  - Push buttons
  - Interrupt signals
  - UART receive
  - Signals between clock domains
  - PLL-generated clocks
- Your system cannot avoid the use of asynchronous inputs.

One or Multiple Clock Domains?

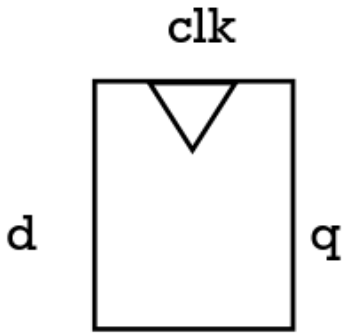
- Examples
  - A clock and its derived clock (via a clock divider) are in the same clock domain because they have a constant phase relationship.
  - CLK, its inversion (derived from CLK) are synchronous to each other.
  - 50MHz and 37MHz clocks (whose phase relationship changes over time) define two separate clock domains.

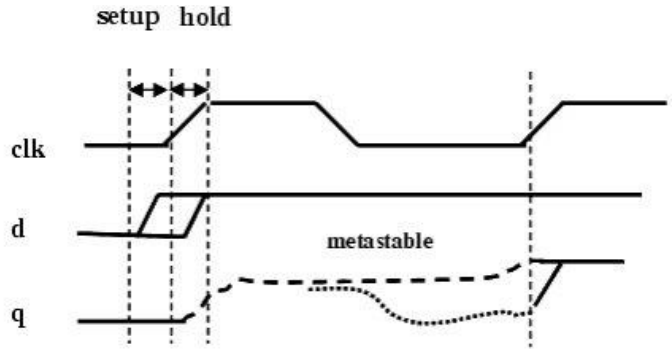


What Is Metastability?

- Flip-Flop Timing Assumptions
  - Setup time (before clock edge)
  - Hold time (after clock edge)
- If setup/hold violated:
  - Internal nodes may not resolve immediately
  - Output may hover between 0 and 1
  - Eventually resolves randomly

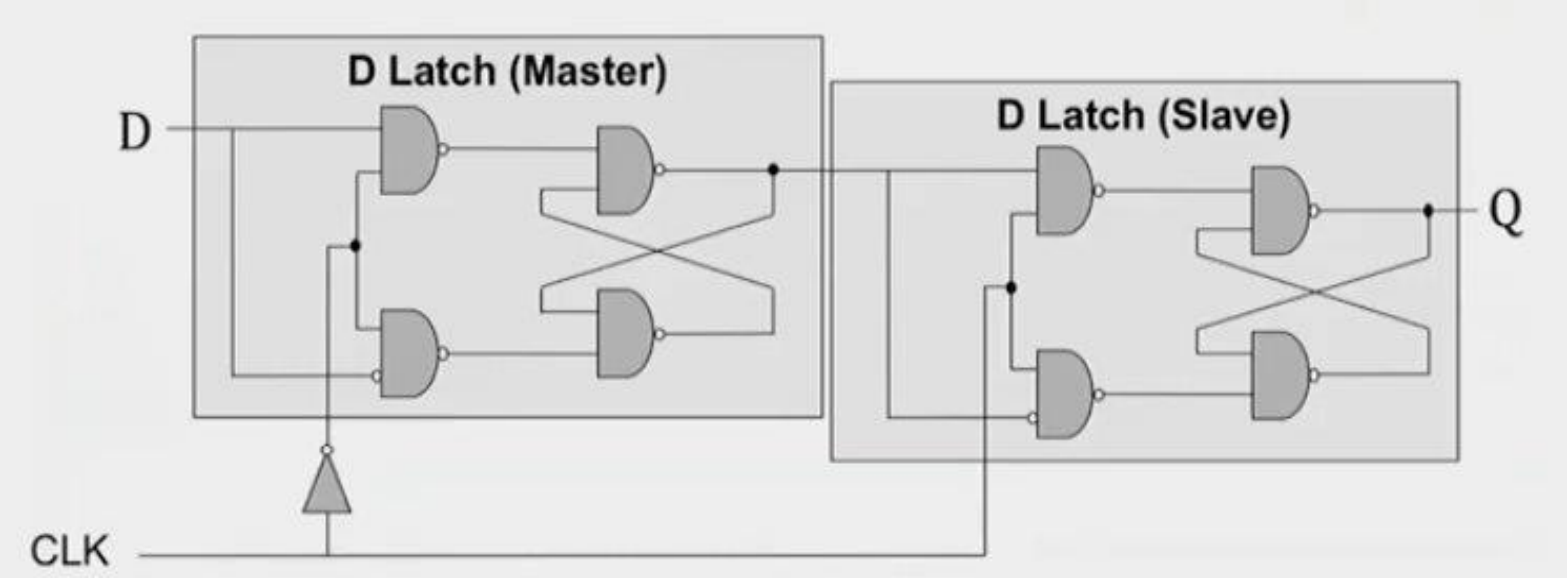
Output is **analog temporarily**
Violates digital abstraction”





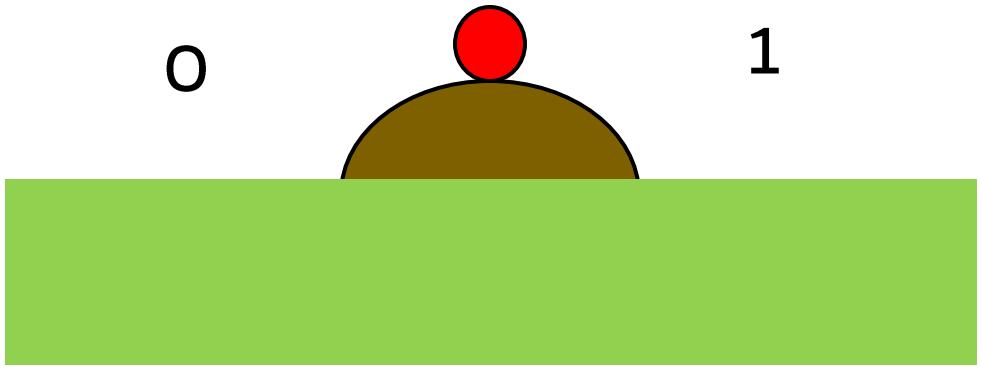
FF Circuit Design

- The clock signal gates D to the cross coupled NAND gates
- D overwrites the state of the first latch when clk is low.
- If D changes when CLK is changing, it's not clear if the overwrite was completed or not.



What Happens Internally, Energy Hill

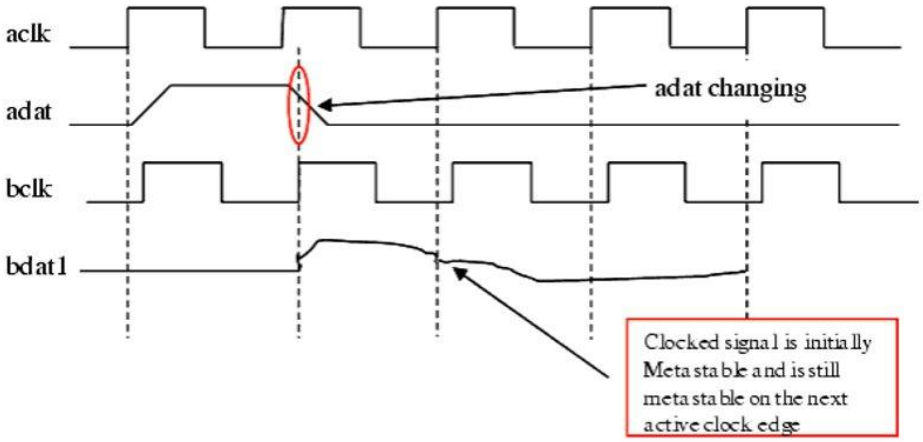
- Imagine:
  - A ball on top of a hill -> internal node voltage
  - It must fall left or right -> logic 0 or 1
  - If perfectly balanced → delay before falling
- Metastability = balanced at the top

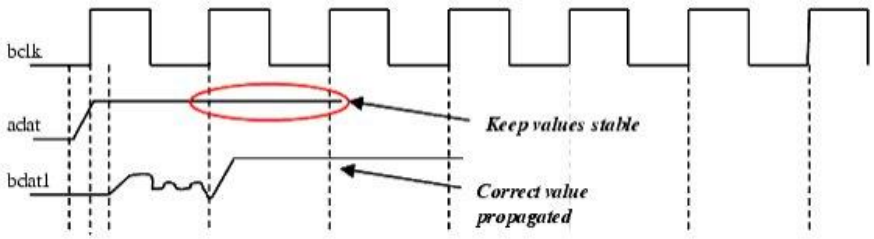


"It cannot be eliminated — only mitigated through design"

Resolution Time

- After a violation:
  - Flip-flop needs time to settle
  - A non-binary input will result in metastability in the next flip-flop
  - If sampled too early → error propagates
- More time = safer
  - Failure probability decreases exponentially with more resolution time.



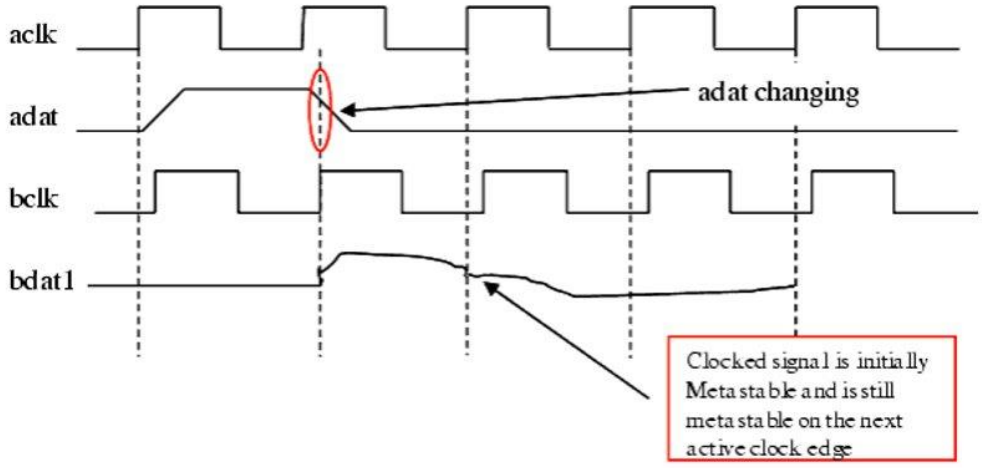


Mean Time Between Failures (Conceptual)

- MTBF depends on:
  - Clock frequency
  - Data toggle rate
  - Available resolution time
- Modern systems are worse:
  - Higher clock frequency → less resolution time
  - Higher data rate → more violations
  - Advanced nodes → smaller margins

Single-Bit CDC Problem

- Clock A → Clock B
  - Sampling occurs at unpredictable phase relationship.
- Risk:
  - Metastability
  - Random 0/1 capture



The Problem: When Digital Assumptions Break

Problem (What is fundamentally wrong?)

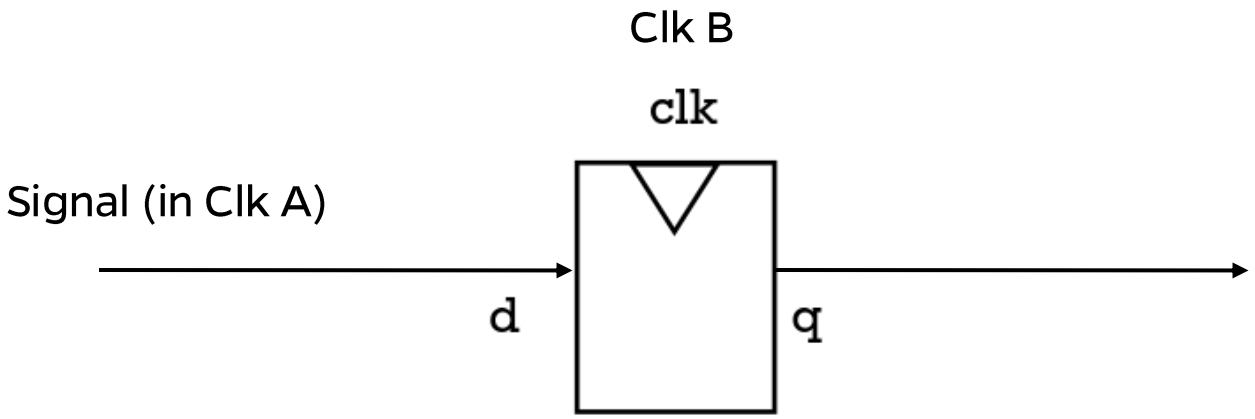
- Digital design assumes:
  - Signals are stable at clock edges
  - Flip-flops output clean 0 or 1
- Reality:
  - Signals can change near the clock edge
  - Setup/hold violations occur
- Root cause:
  - Asynchronous inputs & multiple clock domains (no fixed phase relationship)

Symptoms (What do we observe?)

- Flip-flop output becomes metastable
  - Temporarily analog (not 0/1)
- Output may:
  - Take unbounded time to resolve
  - Resolve to random value
- System-level effects:
  - Non-deterministic behavior
  - Intermittent bugs (hard to debug!)
  - Propagation of errors across logic

Bad Solution

- Directly connecting async signal:



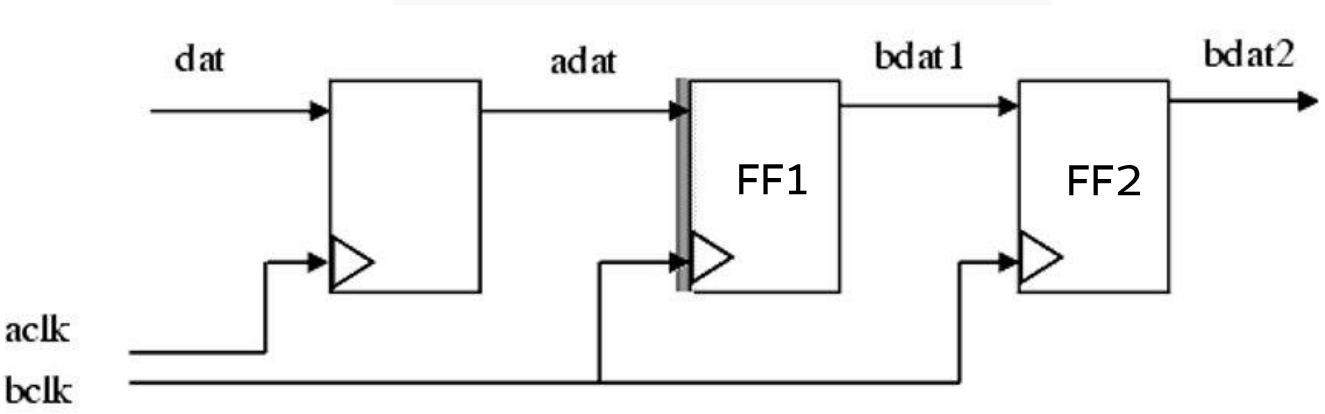
- Problem:
  - Metastable output may propagate
  - Downstream logic corrupted

Dual-Flop Synchronizer

- Example
  - FF1 may go metastable
  - FF2 samples one clock later
  - FF1 has full clock period to resolve
  - Resolution time ≈ one clock cycle

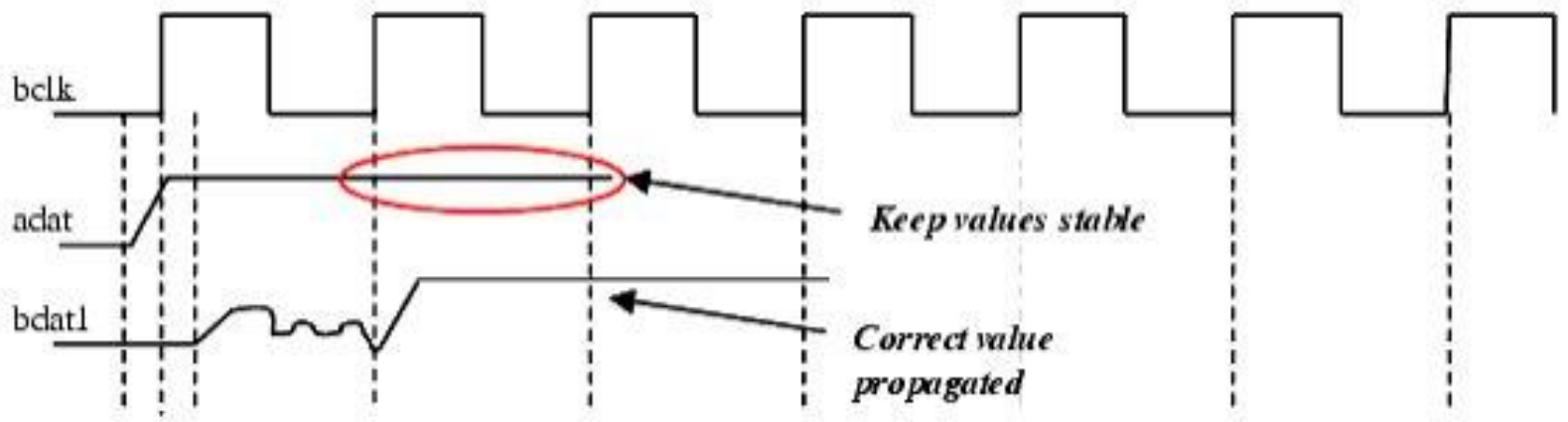
Note that you need to hold the input data for some time.

- MTBF improves exponentially.
  - We are not eliminating Metastability. We are reducing it to an acceptable level



Input Data Stability to avoid Data Loss

- Every transition on the input signal needs to be correctly propagated to the destination domain
- Input signal needs to hold its value a minimum amount of time such that it is sampled
  - Signal must be stable for ≥ 2 destination clock cycles
  - Careful not to miss pulses

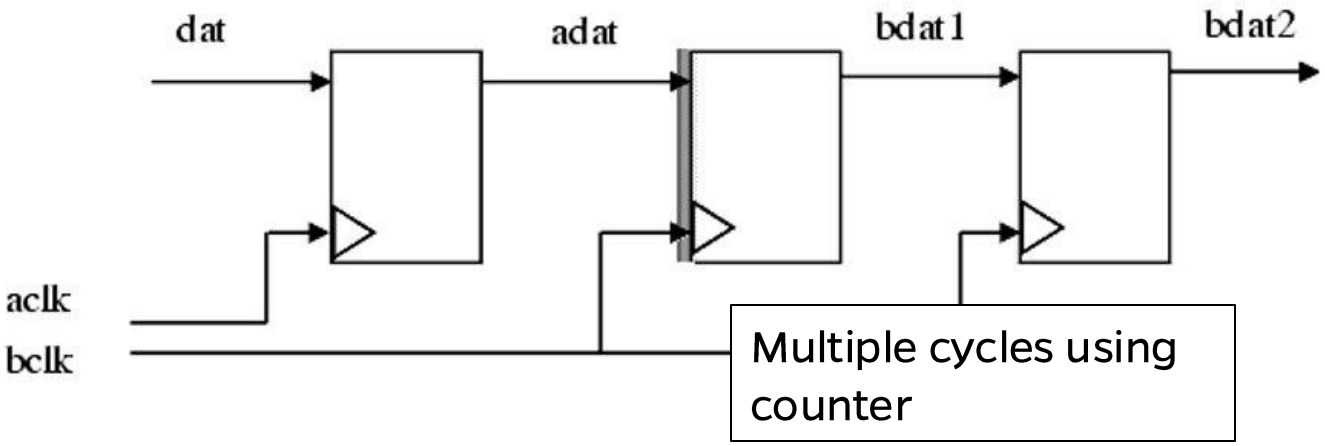


Why Two Flops?

- One flop:
  - Metastability can propagate directly
  - No isolation
  - Failure probability too high
- Two flops:
  - First absorbs metastability
  - Second isolates rest of system
  - Industry standard is two (sometimes three for critical paths)
- Why Not Infinite Flops?
  - Each flop adds latency
  - Diminishing returns

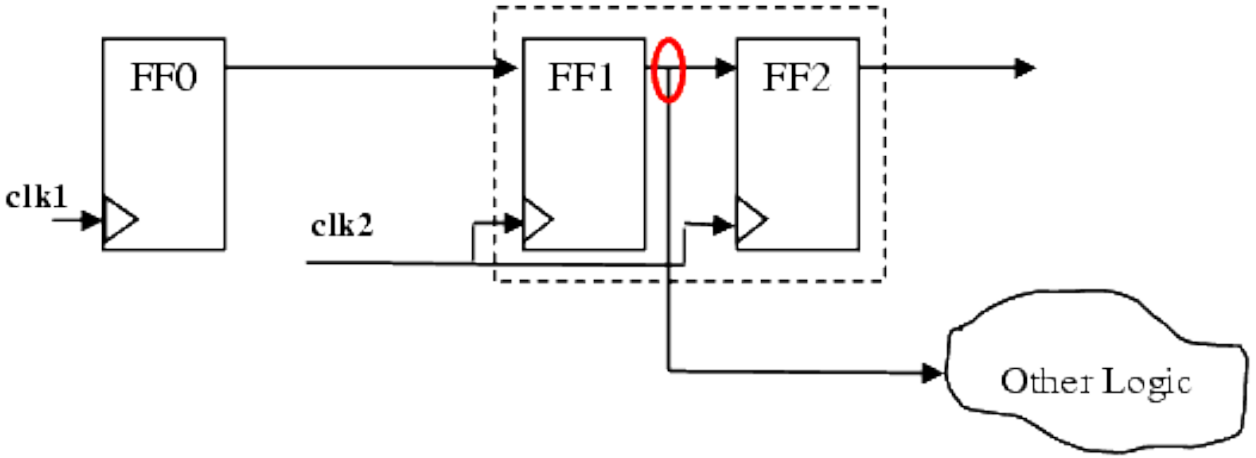
Multi-Cycle Synchronizer Delay

- Add additional delay based on application
- The amount of delay is optimized w.r.t the requirements of MTF



Divergence of Metastable Signal

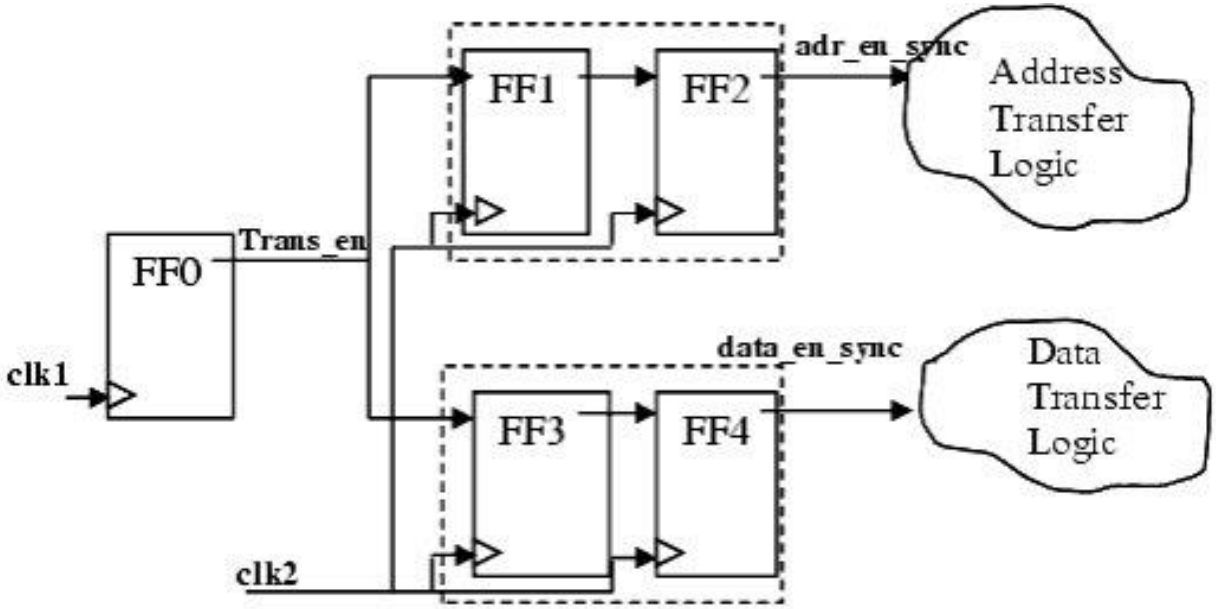
- Using a meta-stable signal in a design can be erroneous. Therefore, multiple fan-out of the output of the first FF is erroneous.



✗ BAD: fan-out of first FF output

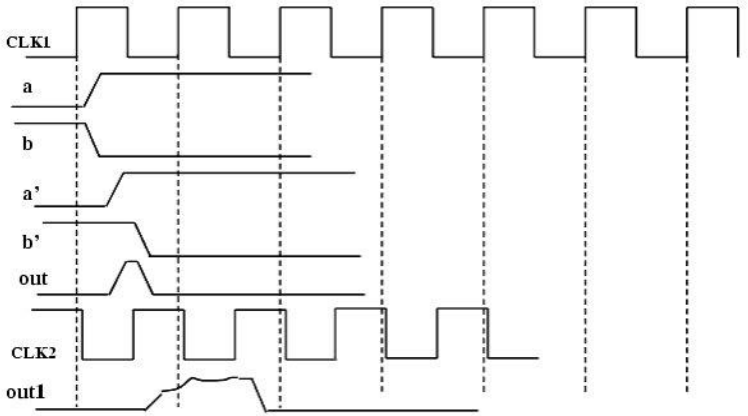
Divergence in the Crossover Path

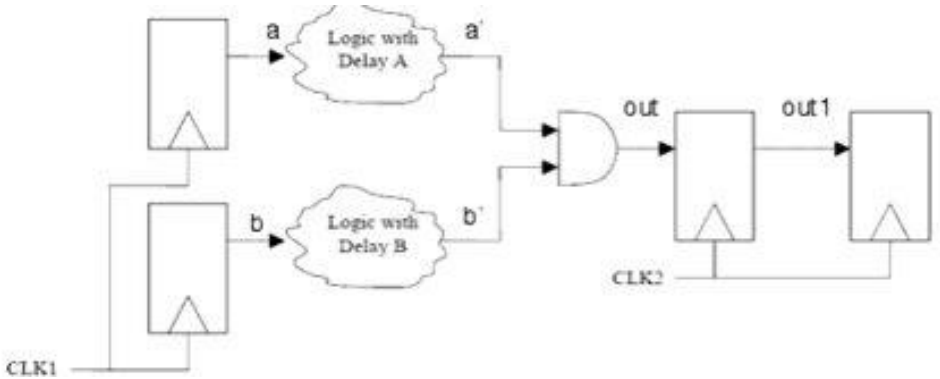
- Design styles which allow divergent logic on CDC signal to multiple synchronization paths



✗ BAD: multiple independent synchronizers for same signal

Convergence in the Crossover Path





What Dual-Flop Does NOT Solve

- Multi-bit data crossing
- Pulse loss
- Bus incoherency
- Throughput issues

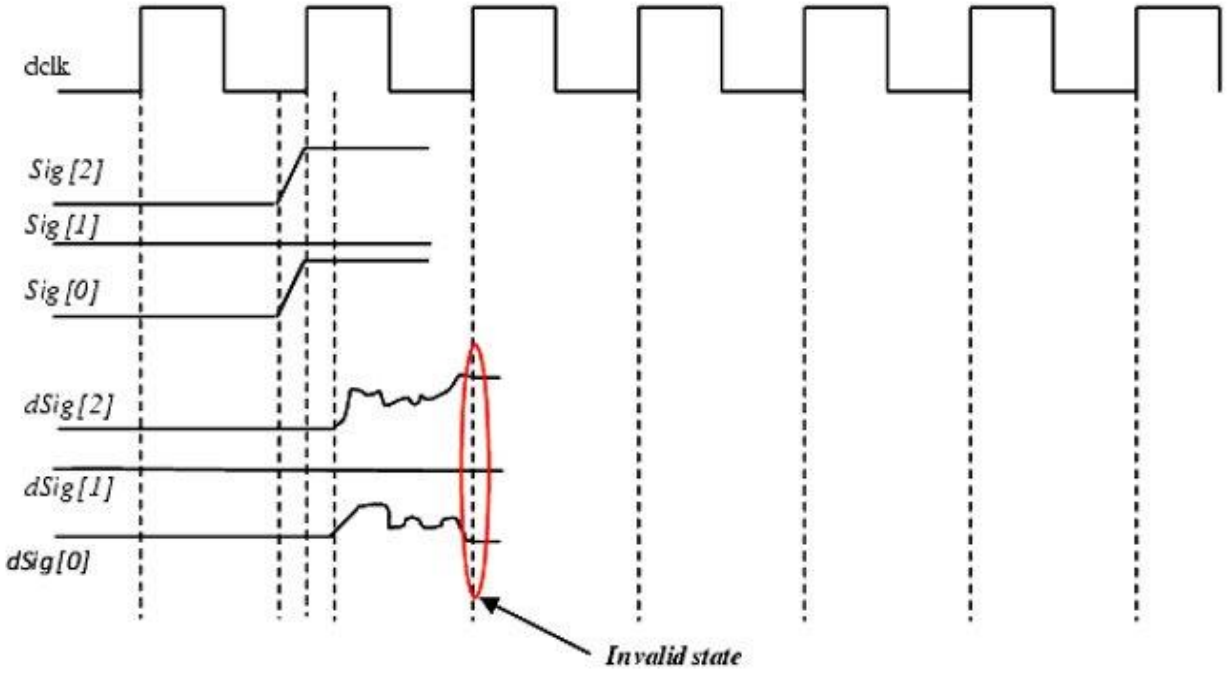
It only protects a **single** control signal!

Failure Case I

- If signal is a short pulse:
  - It may be missed
  - Synchronizer does not stretch pulses
- Important:
  - Synchronizer solves metastability, not protocol.

Failure Case II

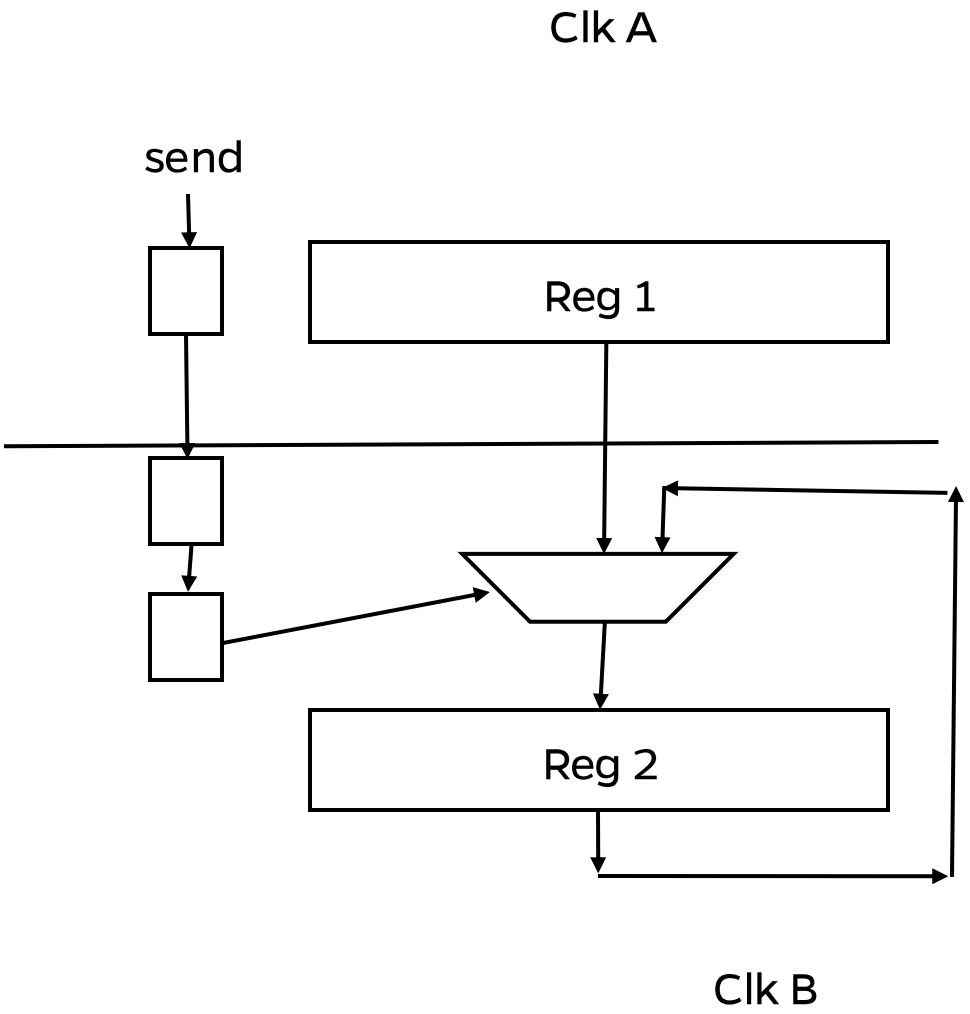
- the natural way to transfer a vector control signal is to model each bit of the vector to be separately synchronized by a FF synchronizer.



Sig wants to change its value from "000" to "101" (both indicate valid states). However, it ends up in invalid state "100".

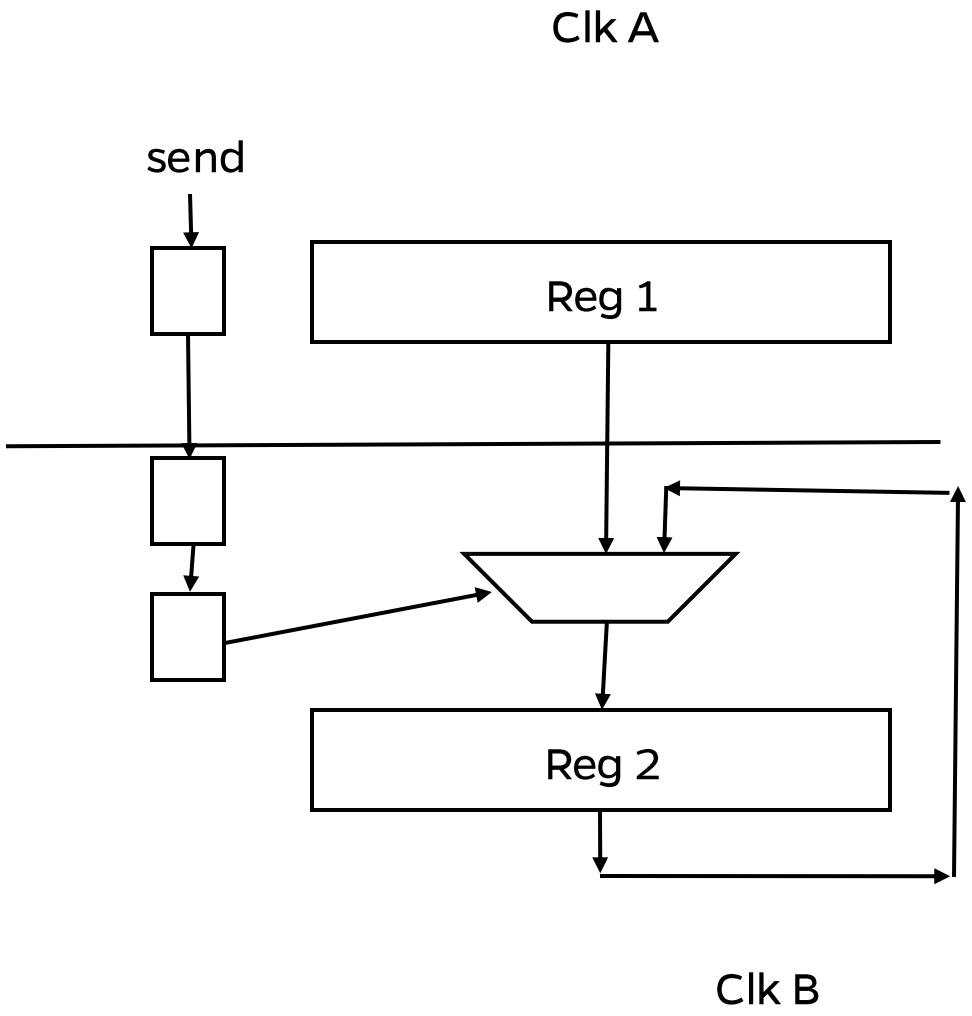
Mux Circulation Register

- Mux controls if new data is accepted or if re-circulated
- Data is applied to reg 1
- Send is held high for a sufficient amount of time for reg 2 to accept the data
- Send is held low for a sufficient amount of time to ensure that the new data loaded in reg 1 is not captured by reg 2 (avoiding metastability).



Mux Circulation Synchronizer

- Limitations:
  - How long is a sufficient amount of time?
  - Low throughput between the clock domains!



Summary

- Metastability is physical and unavoidable
- Violating setup/hold causes it
- Probability reduced with time
- Dual-flop synchronizer mitigates it
- Only valid for single-bit signals